Smart City Bike Sharing System Analysis

The dataset represents a Smart City Bike Sharing system used across several European cities.
It contains information about bike stations, stand capacity, availability, location, and payment options.

Column	Description
Station Name	Name of the bike station
City	City where the station is located
Latitude / Longitude	Geographic coordinates used for map visualization
Bike Stand Capacity	Total number of parking slots available
Available Bikes	Number of bikes currently available
Available Bike Stands	Empty slots available for parking bikes
Payment Type	Whether the station accepts cash or online payment

Data Cleaning Process

- Change Formatting
- Removing null or incomplete records
- Removed un necessary Columns

Before Cleaning

numbe	name	address	position
8265	8265- NÉGRESKO - PAULET	NEGRESKO PAULET - FACE AU 42 RUE NEGRESKO	43.269318556371154, 5.389387542314263
8065	8065-PROMENADE POMPIDOU PALM BEACH	PROMENADE POMPIDOU PALM BEACH - 6 PROMENADE GEORGES POMPIDOU	43.267272515671856, 5.372088524255505
1011	01011 - BORNE TEST NANTES 1	Borne test 1	47.19504, -1.5575
4342	4342 - FLAMMARION MONTRICHET	FLAMMARION MONTRICHET - PLACE LE VERRIER ANGLE BD FLAMMARION ET BD MONTRICHET	43.305993453766845, 5.393678259953781
6176	6176-CANTINI ROUET	CANTINI ROUET - FACE AU 27 AVENUE JULES CANTINI	43.284775983612235, 5.385593763617925
3322	3322 - ARENC CHANTERAC	32 rue de Chanterac 13003 Marseille	43.3123370339008, 5.3687418043728
1264	1264 - CANEBIÈRE DUGOMMIER	CANEBIERE DUGOMMIER - LA CANEBIERE ANGLE BOULEVARD DUGOMMIER	43.29789631329341, 5.381132215559973
1191	1191-CANEBIERE-ST FERREOL	CANEBIERE SAINT FERREOL - ANGLE CANEBIERE SAINT FERREOL	43.29620767061392, 5.376930662205 96
5058	5058 -SAKAKINI - SAINT PIERRE	SAKAKINI SAINT PIERRE - BOULEVARD SAKAKINI ANGLE RUE SAINT PIERRE	43.292800863545835, 5.401285286456447
4293	4293- BLANCARDE	BLANCARDE - GARE DE LA BLANCARDE	43.29626340517055, 5.406000803377506
1245	1245-GAMBETTA - LAFAYETTE	GAMBETTA LAFAYETTE - 6 ALLEES LEON GAMBETTA	43.29902793826475, 5.381500046726694
7153	7153-CORSE ST MAURICE	CORSE SAINT MAURICE - 28 AVENUE DE LA CORSE ANGLE RAMPE SAINT MAURICE	43.28946760889984, 5.363709907863 61
6154	6154-PIERRE PUGET BRETEUIL	PIERRE PUGET BRETEUIL - COURS PIERRE PUGET ANGLE RUE BRETEUIL	43.290329770206334, 5.375548341850035
1002	1002-HOTEL DE VILLE	HOTEL DE VILLE - Face au 42 QUAI DU PORT	43.2961387765145, 5.3707713851323
1224	1224-PLAINE - BIBLIOTHEQUE	PLAINE BIBLIOTHEQUE - 45 RUE DE LA BIBLIOTHEQUE	43.29582518327951, 5.385996090720193
1287	1287- STALINGRAD - RÉFORMÉS	STALINGRAD REFORMES - SQUARE STALINGRAD	43.29952097558015, 5.385047448641554
500	500 - IT PLAISIR		48.799828, 1.946855
5220	5220-ROOSEVELT - ORAN	ROOSEVELT ORAN - 53 COURS FRANKLIN ROOSEVELT ANGLE RUE D'ORAN	43.29899195650368, 5.388799115293999
6160	6160-153 PARADIS	153 PARADIS - 153 RUE PARADIS	43.28683835122117, 5.379450180172236

After Cleaned Up

Bike Station Name	address	position	banking	bonus	status	City	Bike Stands	Available Bike Stands
Novembre	FACE ENTRÉE DU PARC DE LA TÊTE D'OR**	45.779702, 4.860029	FALSE	FALSE	OPEN	Lyon	35	1
Gaston Berger	AVENUE DU 11 NOV./ANGLE AV GASTON BERGER	45.779066, 4.871581	FALSE	FALSE	OPEN	Lyon	35	0
Novembre Négogousses	RUE DU 11 NOVEMBRE FACE 45 RUE DU 14 JUILLET	43.593052, 1.414659	FALSE	FALSE	OPEN	Toulouse	18	6
Liberation	151 LIBERATION - 151 BOULEVARD DE LA LIBERATION	43.30143187524121, 5.392179383331833	TRUE	TRUE	CLOSED	Marseille	10	10
Paradis	153 PARADIS - 153 RUE PARADIS	43.28683835122117, 5.379450180172236	TRUE	FALSE	OPEN	Marseille	14	0
Maisons Charles V (Cb)	RUE DU FAUBOURG DES 3 MAISONS - ANGLE BOULEVARD CHARLES V	48.701365, 6.176824	TRUE	FALSE	OPEN	Nancy	23	23
Mazargues	318 MAZARGUES - FACE AU 318 AVENUE DE MAZARGUES	43.26069960911791 4, 5.393973899334583	TRUE	FALSE	OPEN	Marseille	15	0
Michelet	391 MICHELET - 391 BOULEVARD MICHELET	43.25402727813068, 5.401873594694653	TRUE	FALSE	CLOSED	Marseille	19	17
Prado	397 PRADO - 397 AVENUE DU PRADO	43.26869900851042, 5.385552216536626	TRUE	FALSE	CLOSED	Marseille	13	10
Racine	NOT AVAILABLE	45.764489, 4.877894	FALSE	FALSE	OPEN	Lyon	30	28
Août	ANGLE RUE ROLLET/BLANQUI	45.764812, 4.884311	FALSE	FALSE	OPEN	Lyon	15	12
Août / Pressensé	COURS EMILE ZOLA (VILLEURBANNE)	45.76611, 4.898539	FALSE	FALSE	OPEN	Lyon	19	10
Atelier Vélo'V	NOT AVAILABLE	45.780403, 4.904869	TRUE	FALSE	OPEN	Lyon	3	1
Septembre (Cb)	PLACE DU 8 SEPTEMBRE	47.237686, 6.024183	TRUE	FALSE	OPEN	Besancon	15	9
Corniche Kennedy	88 CORNICHE KENNEDY - 88 CORNICHE J.F. KENNEDY	43.284213979400505, 5.349696475467313	TRUE	FALSE	CLOSED	Marseille	15	15
Libération	91 LIBERATION - 91 BOULEVARD DE LA LIBERATION	43.30050650960441, 5.389298285180646	TRUE	TRUE	CLOSED	Marseille	10	10
De-Octubre-Cieza	NUEVE DE OCTUBRE - CIEZA	39.472098, -0.405361	FALSE	FALSE	OPEN	Valence	20	16
A Rilspert	RUE DES PLATANES	49.60751, 6.045842	FALSE	FALSE	OPEN	Luxembourg	15	7
A Schommesch	RUE D'ERNSTER 18	49.659778, 6.241714	FALSE	FALSE	OPEN	Luxembourg	20	14

Dax Measures

```
AvgAvailableBikes = AVERAGE(Bike[Available Bikes])
Bike Stand Capacity = SUM(Bike[Bike Stands])
StationCount = COUNTROWS(Bike)
```

Dax Custom Column

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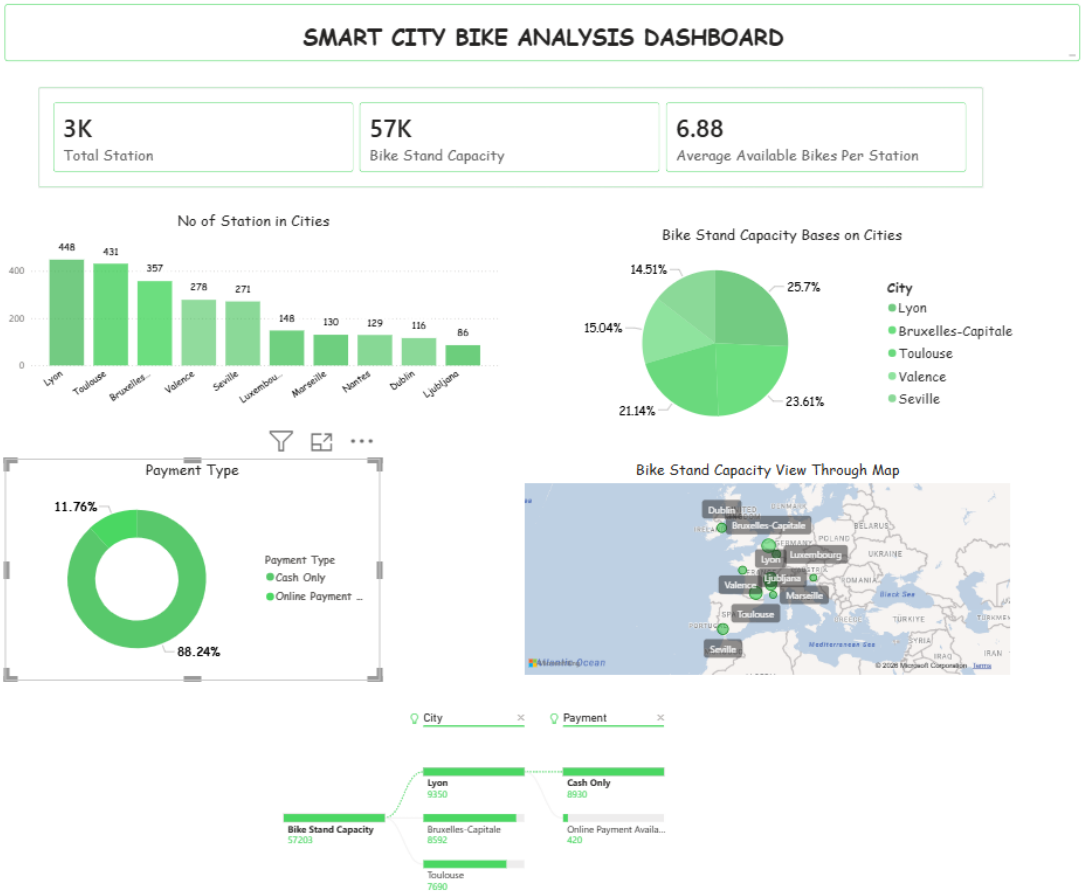
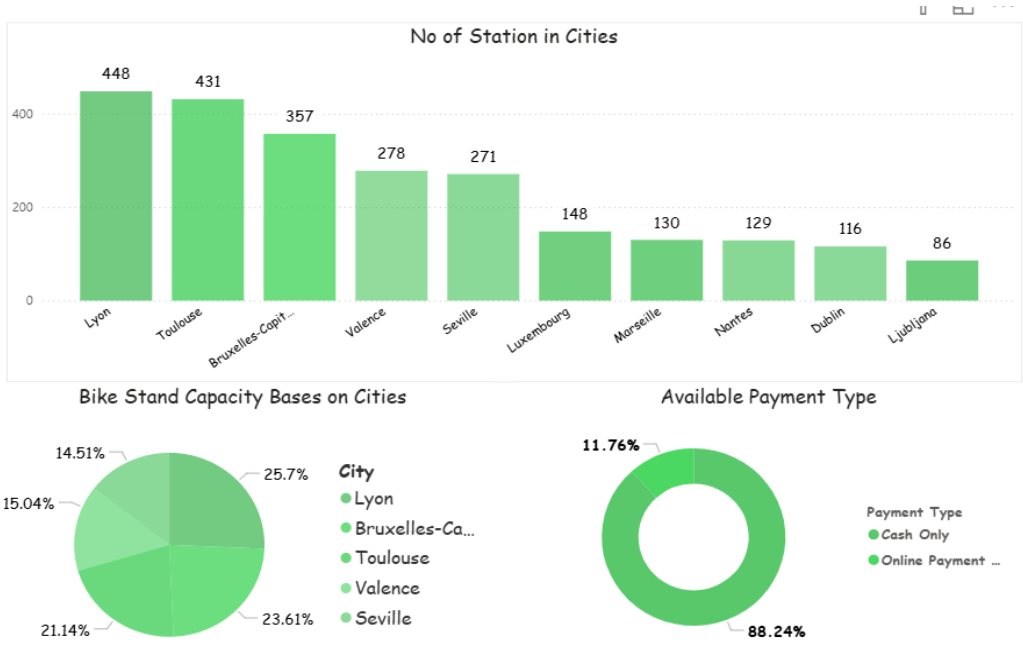
Payment = IF(Bike[banking] = True,"Online Payment Available","Cash Only")
```



The dashboard analyzes the bike-sharing infrastructure across multiple European cities to understand the availability of bike stations, stand capacity, and payment accessibility.

- The network contains approximately 3,000 bike stations across cities.
- The total bike stand capacity is around 57,000, indicating the infrastructure scale of the bike-sharing system.
- On average, each station has 6.88 available bikes, showing moderate availability for commuters.



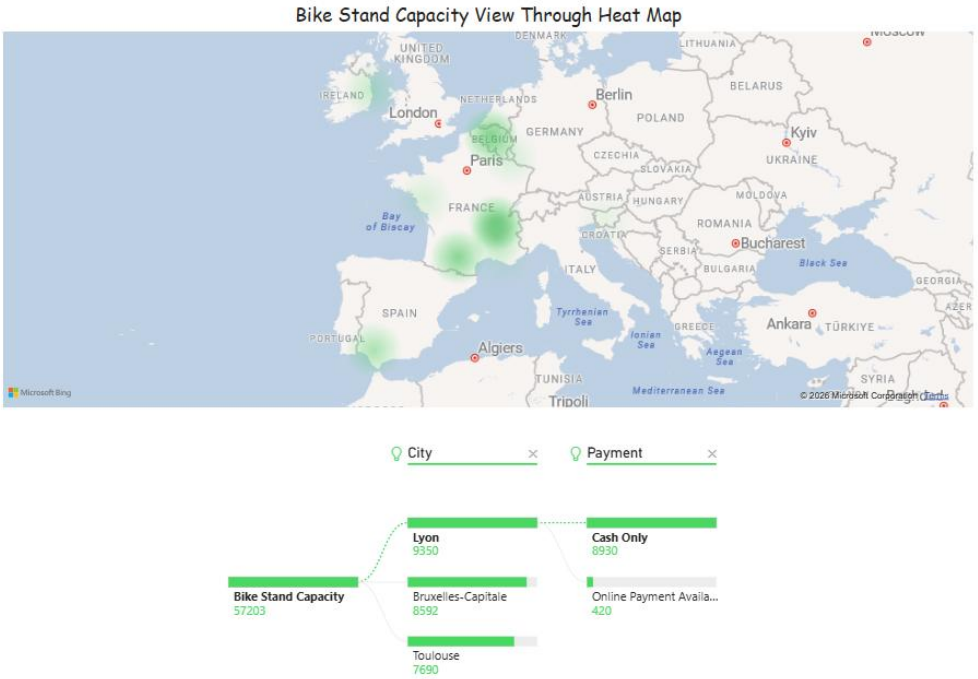
City Distribution

- Lyon has the highest number of stations (448 stations).
- Toulouse follows with 431 stations.
- Brussels-Capital has 357 stations.
- Smaller networks exist in cities like Dublin and Ljubljana.

This indicates that bike-sharing adoption is stronger in certain urban areas compared to others.

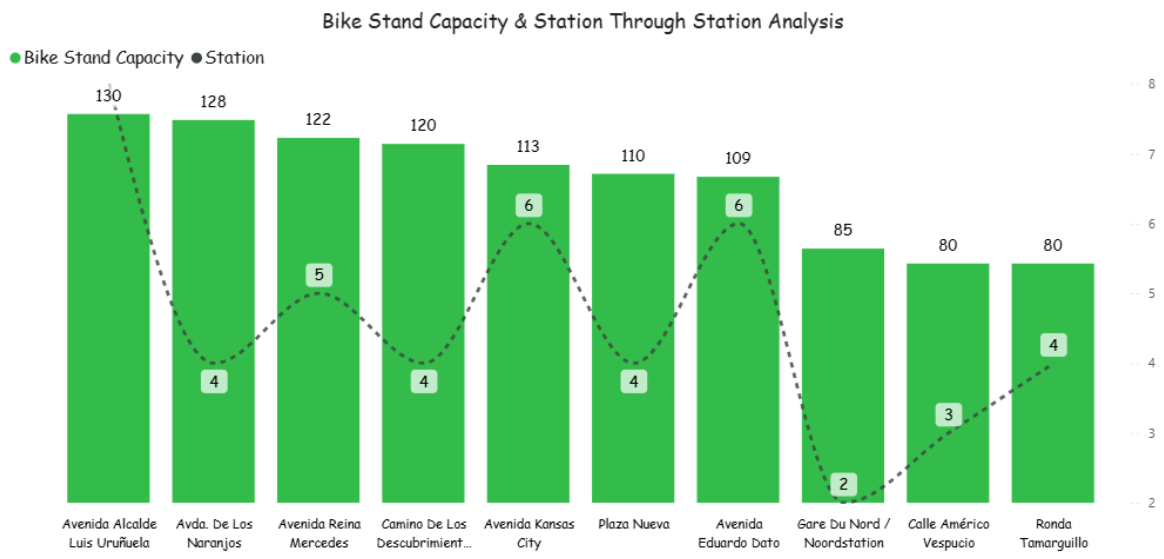
Capacity Distribution

- Lyon contributes about 25.7% of total stand capacity, making it the dominant city in infrastructure.
- Toulouse and Brussels-Capital also hold significant shares.
- Valence and Seville have smaller capacity contributions.



Payment Accessibility

- 88.24% of stations accept cash-only payments
- Only 11.76% support online payments



Predictive Analysis

- Cities with high station counts (Lyon, Toulouse) will likely experience increasing demand for bike availability and stand expansion.
- If urban mobility trends continue, cities with fewer stations may see infrastructure growth.
- The low online payment adoption (11.76%) suggests strong potential for digital payment expansion.
- Bike shortages may occur during peak hours
- Station congestion could increase in high-demand cities

Prescriptive Analysis

Optimize Station Capacity

- Stations with very high stand capacity but lower station density should be evaluated to ensure bike redistribution efficiency.

Expand Networks in Underrepresented Cities

- Cities like Dublin, Marseille, and Ljubljana have lower station counts and could benefit from network expansion.

Expand Digital Payments

Since most stations are cash-only, introducing more online payment systems and mobile app payments could:

- Improve user convenience
- Increase system usage
- Reduce operational friction

Conclusion

By expanding station networks, improving digital payment adoption, and balancing infrastructure distribution, cities can significantly improve **urban mobility,** sustainability, and commuter accessibility.